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Lab Practical #02:

Study of different network devices in detail.

Practical Assignment #02:

1. Give difference between below network devices.
   - Hub and Switch
   - Switch and Router
   - Router and Gateway
2. Working of below network devices:
   - Repeater
   - Modem((DSL and ADSL)
   - Hub
   - Bridge
   - Switch
   - Router
   - Gateway

Hub and Switch

No.	Hub	Switch
1	Sends data to all connected devices, which causes more network traffic.	Sends data only to the specific device it is meant for.
2	Works at a slower speed due to unnecessary data transmission.	Works at a faster speed by reducing unnecessary data sharing.
3	It is not smart, does not store device information.	It is smart, stores MAC addresses to identify devices.
4	Less secure, as data goes to all devices.	More secure, as data is sent only to the target device.
5	Rarely used in modern networks.	Commonly used in modern networking setups.

Switch and Router

No.	Switch	Router
1	Connects devices within the same network like LAN.	Connects different networks like LAN to WAN.
2	Sends data based on MAC addresses.	Sends data based on IP addresses.
3	Works faster for local data transfer.	Works slower due to complex routing tasks.

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4	Cannot assign IP addresses to devices.	Can assign IP addresses using DHCP.
5	Used mainly in internal networks.	Used to connect to external networks like the internet.

Router and Gateway

No.	Router	Gateway
1	Connects two or more networks (usually similar types).	Connects different types of networks (Exa., LAN to cloud).
2	Works at the network layer (Layer 3).	Works at all layers of the OSI model if needed.
3	Uses IP addresses to route data.	Translates protocols between networks.
4	Mostly used in home and office networks.	Used in large or complex systems
5	Faster and simpler in function.	Slower due to protocol conversion tasks.

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Working of below network devices:

1. **Repeater**
   A repeater receives a weak signal and regenerates it to extend the distance of a network.
   It works at the physical layer (Layer 1) of the OSI model.

2. **Modem (DSL and ADSL):**
   A modem converts digital signals to analog and vice versa for internet over phone lines.
   DSL/ADSL modems allow simultaneous use of internet and telephone.

3. **Hub:**
   A hub broadcasts data to all connected devices regardless of destination.
   It works at the physical layer and does not filter traffic.

4. **Bridge:**
   A bridge connects two similar networks and filters traffic based on MAC addresses.
   It helps reduce traffic by dividing a network into segments.

5. **Switch:**
   A switch sends data only to the intended device using MAC addresses.
   It operates at the data link layer and improves network efficiency.

6. **Router:**
   A router connects different networks and routes data based on IP addresses.
   It works at the network layer and is commonly used to access the internet.

7. **Gateway:**
   A gateway connects networks using different protocols and acts as a translator.
   It operates across multiple OSI layers to enable communication between systems.